`extended_stats` aggregations produce multiple values as output. For example, the `stats` aggregation output consists of `min`, `max`, `sum`, `avg`, and a couple more for the same set of documents.

Most metric aggregations are self-explanatory. For instance, a `sum` aggregation sums up all the given values, and `avg` averages the values. If you need to work with metrics not discussed in this chapter, see the documentation on the Elasticsearch site for their usage: http://mng.bz/qrRz.

13.2.1 Sample data

In the next few sections, we discuss popular aggregations. Before we do that, let's prime Elasticsearch with a few documents. The following listing primes our data store with a list of TV sales by creating a new index, `tv_sales`. You can also download the same sample set from GitHub (http://mng.bz/7DQ4) or the book's website.

Listing 13.1 Indexing TV sales data

```
PUT tv_sales/_bulk
{"index":{"_id":"1"}}
{"brand": "Samsung","name":"UHD TV","size_inches":65,"price_gbp":1400,
➥ "sales":17}
{"index":{"_id":"2"}}
{"brand":"Samsung","name":"UHD TV","size_inches":45,"price_gbp":1000,
➥ "sales":11}
{"index":{"_id":"3"}}
{"brand":"Samsung","name":"UHD TV","size_inches":23,"price_gbp":999,
➥ "sales":14}
{"index":{"_id":"4"}}
{"brand":"LG","name":"8K TV","size_inches":65,"price_gbp":1499,"sales":13}
{"index":{"_id":"5"}}
{ "brand":"LG","name":"4K TV","size_inches":55,"price_gbp":1100,"sales":31}
{"index":{"_id":"6"}}
{"brand":"Philips","name":"8K TV","size_inches":65,"price_gbp":1800,
➥ "sales":23}
{"index":{"_id":"7"}}
{"name":"8K TV","size_inches":65,"price_gbp":2000,"sales":23}
{"index":{"_id":"9"}}
{"name":"8K TV","size_inches":65,"price_gbp":2000,"sales":23,
➥ "best_seller":true}
{"index":{"_id":"10"}}
{"name":"4K TV","size_inches":75,"price_gbp":2200,"sales":14,
➥ "best_seller":false}
```

This listing indexes documents with various attributes to a `tv_sales` index. Take particular note of the `best_seller` field: it is only set on the last two records. Now that we have a sample data set, let's run a few common metric aggregations.

13.2.2 The value_count metric

The `value_count` metric counts the number of values present for a field in a set of documents. If our requirement is to fetch the number of values in Elasticsearch for a given